7 Bridges, a type of structure commonly found in daily life, are often built to span over physical obstacles such as a body of water or a valley, for the purpose of providing passage over the obstacles.

Designs of bridges depend mainly on the function of the bridge, the nature of the terrain where the bridge is constructed and anchored, the material used to make it, and the funds available to build it. Common types of bridges are beam, truss, cantilever and suspension bridges. A beam bridge, support by pillars at its both ends, span across a stream, is illustrated in Fig. 7.1.

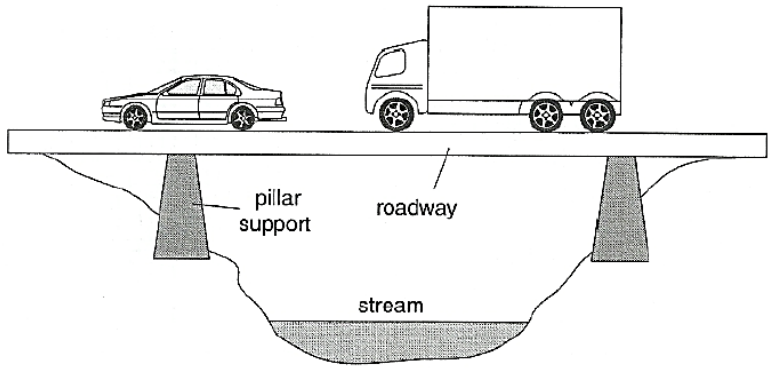

Fig. 7.1

The traditional building materials for bridges are stones, wood and steel, and more recently reinforced and pre-stressed concrete. These materials have different qualities of strength, workability, durability and resistance against corrosion. They differ also in their structure, texture and colour. For bridges, one should use a material which results in the best bridge in terms of shape, technical quality, functionality, economics and compatibility with the environment.

(a) Bridges bend when vehicles or pedestrians cross them. Hence, it is important to ensure that bridges do not bend too much when they are loaded.

Fig. 7.2 shows a simple model of a beam bridge built to investigate how the deflection of a wooden beam varies with the amount of load applied to the beam. The beam of length L, is supported on its two ends. A load of weight W is hung from the center of the beam. The deflection d is measured for the different loads and length of beams. A series of graphs are plotted as shown in Fig. 7.3.

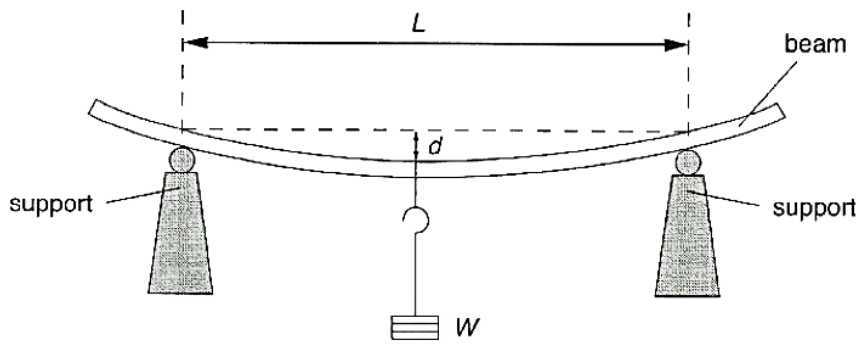

Fig. 7.2

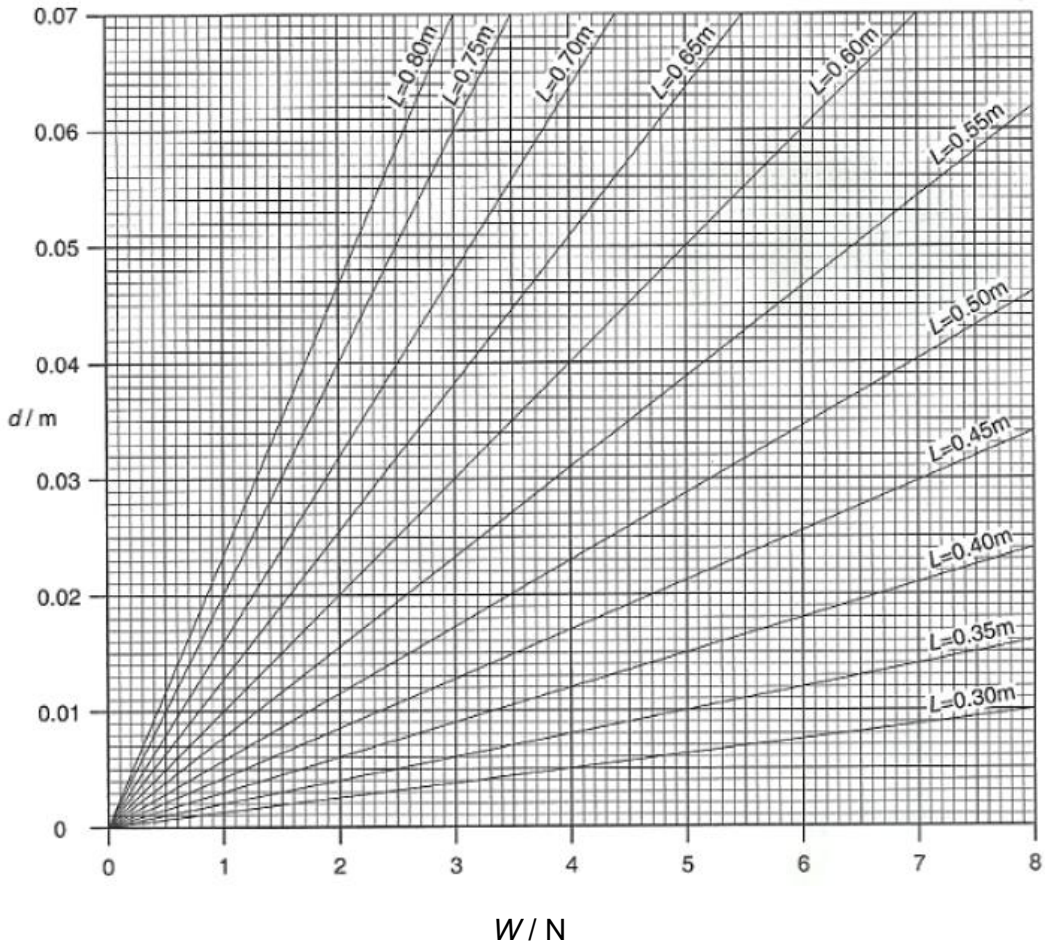

Fig. 7.3

The relation between d and L is thought to follow the expression

$$d = k\,L^n$$

where k and n are constant.

(i) A load of 3 N is applied to the different length of beams. [3]

Complete Fig. 7.4.

L / cm	d / cm	log (L /cm)	log (d /cm)
30	0.40	1.48	-0.40
35			
40	0.90	1.60	-0.046
45			
50	1.70	1.70	0.230
55	2.30	1.74	0.362
60	3.00	1.78	0.477
65			
70	4.80	1.85	0.681
75	6.00	1.88	0.778

Fig. 7.4

(ii) Plot a graph of log (d / cm) against log (L / cm) in Fig. 7.5. [3]

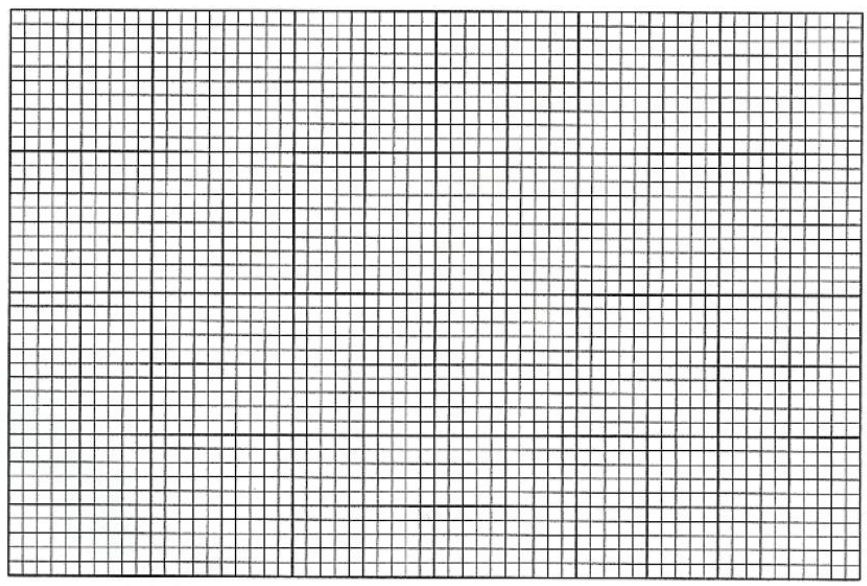

Fig. 7.5

(iii) Using Fig. 7.5, determine the value of k. The unit of k is not required.

k = …….. [2]

(b) A beam bridge supported at both of its ends might also sag under its own weight. Hence the stiffness of the materials used to build the bridges is important. A physical quantity, Young's modulus, Y, also known as elastic modulus, is a measure of stiffness of material. It is the ratio of tensile stress to tensile strain exerted on the material, i.e.

$$Y = \text{Tensile stress / Tensile strain}$$

To measure the Young's modulus of the wooden beam in Fig. 7.2, forces F of same magnitude but opposite direction is applied on both ends of a wooden beam of length Lo = 1.000 m and cross-sectional area A of 5.0×10^{-5} m^2 as shown in Fig. 7.6. Under tension forces, the beam will elongate. Its extension in length is denoted as ΔL.

The tensile stress of the wooden beam is the ratio of the force F to the cross-sectional area A. While the tensile strain of the wooden beam is the ratio of the extension of length ΔL to the original length L.

$$\text{Tensile stress} = F / A$$

$$\text{Tensile strain} = \Delta L / L$$

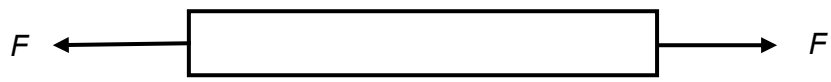

Fig. 7.6

Fig. 7.7 shows the forces applied to the beam and the respective extension of the beam.

F / N	ΔL / mm
1.0	0.018
2.0	0.032
3.0	0.060
4.0	0.072
5.0	0.092
6.0	0.112

Fig. 7.7

(i) Plot a graph of force F against extension ΔL in Fig. 7.8. [3]

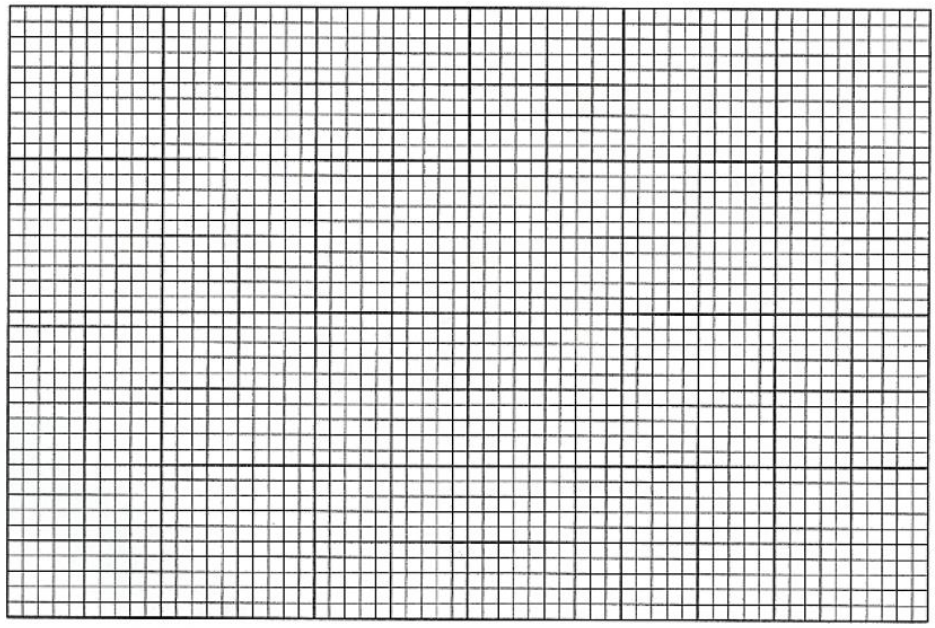

Fig. 7.8

(ii) From Fig. 7.8, determine the gradient of the graph.

gradient = …….. [2]

(iii) Hence or otherwise, determine the Young's Modulus, Y, of the wooden beam.

Y = ……..N mm^{-2} [2]

(iv) A structural engineer wants to use a stiffer material instead of wood to construct a beam bridge.

1. Suggest a material which is stiffer than wood and can be used to construct a beam bridge.

……... [1]

2. In Fig. 7.8, sketch the force against extension graph of a beam made of a stiffer material than wood. Label the graph S. Explain your answer.

……

……

…… [2]

(c) The loading of the beam is often non-uniform along its length, as shown in Fig. 7.9. This means that the moments of forces have to be considered.

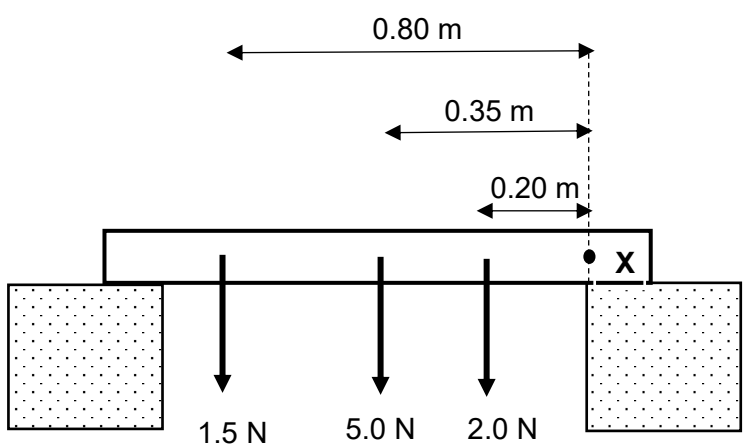

Fig. 7.9

Calculate the total moment of the forces acting on the beam about point X in Fig. 7.9. You may assume that the weight of the beam is negligible.

moment = ……..N m [2]

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